

# Joseph Wallace

Computational Chemist | Molecular Modelling | Ligand Screening Workflows

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## PROFESSIONAL SUMMARY

Computational chemist with a background in molecular modelling, simulation, and structure-focused analysis across biomolecular and nanoscale systems. Currently building reproducible ligand-screening, docking, rescoring, and scientific Python workflows for drug-discovery-style prioritisation. Strong focus on physically grounded modelling, benchmark-aware evaluation, and clear computational decision pipelines.

## CORE SKILLS

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|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Molecular Modelling &amp; Screening</b> | Docking workflows, rescoring strategy, molecular interaction analysis, structure-based modelling, comparative ranking and benchmark evaluation |
| <b>Scientific Computing &amp; Analysis</b> | Python, NumPy, pandas, SciPy, matplotlib; reproducible analysis pipelines, workflow automation, data visualisation                             |
| <b>Simulation Methods</b>                  | Molecular dynamics, enhanced sampling, umbrella sampling, metadynamics, Hamiltonian replica exchange, free-energy-oriented analysis            |
| <b>Cheminformatics &amp; Tools</b>         | RDKit, PyMOL, VMD, xTB                                                                                                                         |
| <b>HPC &amp; Research Computing</b>        | Linux environments, HPC workflows, job submission, simulation pipeline management                                                              |

**Tools:** Python • RDKit • GROMACS • PLUMED • PyMOL • VMD • xTB • Linux/HPC

## PROFESSIONAL EXPERIENCE

### Postdoctoral Researcher

2022–Present

University of Padova / Italian Institute of Technology — Italy

- Lead computational modelling and analysis on nanoparticle–molecule systems, with emphasis on physically grounded simulation, molecular interaction analysis, and reproducible scientific workflows.
- Designed and maintained Python-based analysis pipelines for simulation data, including automated processing, visualisation, and benchmark-style comparison across systems.
- Applied molecular simulation and enhanced-sampling methods to study structure, dynamics, and interaction behaviour in complex molecular systems.
- Worked across interdisciplinary settings, translating molecular-scale results into interpretable hypotheses and decision-ready outputs.

### PhD Research Attachment

2021–2022

Université Sorbonne Paris Nord — France

- Supported research continuity during laboratory relocation, including onboarding, coordination, and integration into collaborative structures.
- Contributed to computational and organisational research activity within an international academic environment.

### PhD Research Attachment

2019–2021

A\*STAR Institute of High Performance Computing — Singapore

- Conducted computational biochemistry research in an HPC environment, focusing on molecular simulation, structural analysis, and data-intensive modelling workflows.
- Worked in an international setting with strong emphasis on scalable computation and reproducible analysis.

## EDUCATION

### PhD, Computational Biochemistry

2017–2022

University of Liverpool — UK

*Thesis:* How does constraining folded peptides within dense self-assembled monolayers on gold nanoparticles affect their structure?

- Combined molecular simulation and experimental data to study structure, dynamics, and interactions in peptide–nanoparticle systems.

- Built a strong foundation in molecular modelling, structural interpretation, and computational analysis.

### **BSc Biochemistry, First Class Honours**

2014–2017

University of Liverpool — UK

*Project:* Solving the crystal structure of Cytochrome bc<sub>1</sub> for the development of antimalarials.

### **Summer Studentship (Wellcome Trust Funded)**

Summer 2016

University of Liverpool — UK

*Project:* Investigating the potential of non-heparin anticoagulants with antithrombin.

## **SELECTED PROJECTS**

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### **Docking + Rescoring Benchmark for Ligand Ranking**

- Built a reproducible workflow to test whether rescoring improves prioritisation over baseline docking on a curated protein–ligand benchmark.
- Focused on ranking analysis, benchmark design, and explicit failure-mode evaluation rather than inflated performance claims.
- *Links:* [Project page](#) | [Repository](#)

### **Simulation Analysis Pipelines for Molecular Systems**

- Designed Python-based workflows for processing, comparing, and visualising molecular simulation outputs across multiple systems.
- Emphasis on reproducibility, interpretable analysis, and workflow reuse.

### **Low-Data Molecular Modelling Study**

- Exploratory case study comparing baseline molecular representations, transfer-learning ideas, and latent diagnostics under limited-data conditions.
- *Status:* analysis prototype; claims kept deliberately narrow.

## **SELECTED PUBLICATIONS**

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1. Wallace, J., Riccardi, L., Mancin, F., & De Vivo, M. (2025). A Scoring Function for Monolayer-Protected Gold Nanoparticles Capable of Recognizing Small Organic Molecules in Solution. *Journal of Chemical Theory and Computation*, **21**(21), 11070–11078. DOI: 10.1021/acs.jctc.5c01278
2. Franco-Ulloa, S., Cesari, A., Zaroni, G., Riccardi, L., Wallace, J., Mascitti, B. B., Rastrelli, F., Mancin, F., & De Vivo, M. (2025). Rational design of gold nanoparticle-based chemosensors for detection of the tumor marker 3-methoxytyramine. *Chemical Science*, **16**, 6282–6289. DOI: 10.1039/D4SC08758E
3. Wallace, J. (2022). *How Does Constraining Folded Peptides within Dense Self-Assembled Monolayers on Gold Nanoparticles Affect Their Structure? An Experimental and Computational Investigation*. PhD thesis, University of Liverpool.

## **ADDITIONAL EXPERIMENTAL BACKGROUND**

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Gold nanoparticle synthesis and characterisation (UV/Vis, FTIR, fluorescence); structural biochemistry methods including circular dichroism, HPLC, and crystallography exposure.